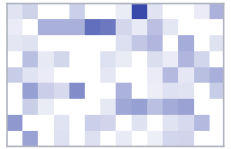
DATA

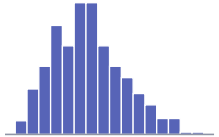
Expression data

single-cell counts
genes × cells



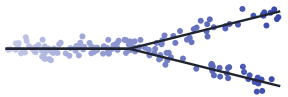
Preprocessing

normalisation,
gene filtering



Pseudotime (optional)

cell ordering, e.g. Slingshot;
needed by time-based methods



INFERENCE

GRN inference

Tree ensembles

GENIE3, GRNBoost2

Information theory

PIDC, SCRIBE

Correlation

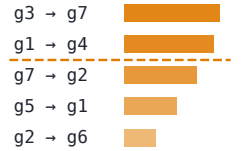
PPCOR, LEAP

Regression / ODE

SCODE, GRISLI, SINCERITIES

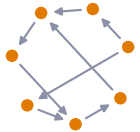
Ranked edge list

score for every
regulator → target pair



Predicted network

top-*k* or score threshold;
edges may be directed/signed



all edges

EVALUATION

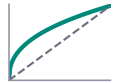
Metrics

Threshold-free

AUROC, AUPR

Top-*k*

early precision, EPR



Reference network

simulated ground truth or
curated gold standard